



Invisible Optical Fibre

Scientists have reportedly found a new way to transmit and amplify light signals without using cables <http://dgit.in/NewOpFibre>



First Arab Mars Mission

The UAE plans to have a mars mission and build an Arab spaceship in 7 years <http://dgit.in/ArabSpaceship>

However, this is not an exact definition, as it encompasses many different jobs descriptions and skills that do not always overlap. For example, a Data Scientist might only run tests on data using enterprise software, another might write a new algorithm to analyse data. Yet another might make inferences based on results from these analyses and spend more time communicating with clients.

This has led to people questioning if we should even be using the term 'Data Science'? According to Ravi Mohan, a Machine Learning programmer based in Bangalore, "Data Science is a fancy name for applied statistics. It is a marketing buzzword...The meaning is too fuzzy to be of any real use." Perhaps it is best not to use the term Data Science at all, and merely use job titles that describe what people in the field do - Programmer, Software Engineer, Data Analyst. But some people find these terms outdated or vague, and Data Science is already a buzzword. For this reason, we will continue using it, but for the purpose of this article, make a distinction between different domains within data science.

Opportunities in Data Science

As mentioned earlier, there is a lot of data coming in, and we need to process and make sense of this. As such, more and more opportunities are being created in data science. In June 2014, NASSCOM projected returns of \$7.5 billion dollars from the big data analytics market by 2020. Advisory company Gartner predicts Big Data related jobs to grow by 4.4 million worldwide by 2015.

In India, the job focus so far has centered on another buzzword - Analytics - and more recently, Big Data Analytics, which more often than not refers to a combination of Data Infrastructure, Business Intelligence and Data Science. Though the pay and career progression in Data Infrastructure and BI can be good in its own right, as discussed earlier, these are different fields of expertise.

Looking only at what can be considered Data Science, the majority of work opportunities available seem to target the low hanging fruit, outsourced work at BPOs and KPOS. Tools used here involve

SAS and SPSS, and you can expect a starting salary of 5 lakhs a year. This is not bad, but you could be getting a lot more if you know how to code.

All the real data science, the cutting edge work, seems to be taking place at smaller organisations and startups that specialise in advanced analytics, like Fractal Analytics, Mu Sigma, Absolut-Data, Manthan, Ananto Analytics and Imagina Analytics. The typical starting salary for someone using a combination of statistical, programming and business knowledge is around 10 lakhs a year. This goes up to 15 lakhs if for those with experience and really good at what they do. A few more years of experience as you grow to be a senior analyst and managing a team can net you 25 lakhs a year.

Desired Skills

Data Science broadly makes use of skills that involve collecting, storing and querying massive amounts of data, analysing it critically using complex computational statistical methods, creating visualisations, coming up with inferences and predictions, and communicating these to others. The two broad types of skills you need are statistical and computational. You need to be able to program better than a typical Statistician, and be able to understand statistics better than the average Computer Scientist.

The major hiring challenge right now for companies in India is finding candidates that possess all these skills. Software engineers are good with different data types and programming, but do not necessarily have detailed knowledge of statistical models, let alone their assumptions, and how to derive inferences from their results. Statisticians are brilliant

at this, but their knowledge is highly theoretical; they lack advanced programming skills and have little experience working with different data types. Also, both Computer Scientists and Statisticians lack business domain knowledge that serves as context to understand a problem. For this reason, Indian analytics companies find it easier to hire people with engineering and MBA degrees who have some programming skills, and train them on statistical concepts.

Most companies in India accept a B.E or MSc in Math, Statistics or Economics, and a PhD would be a plus, though organisations abroad tend to be more flexible, letting in people from other fields as long as they have the requisite programming and statistical skills. And India being India, people with a MBA are preferred. Organisations here want people who understand how various business domains work, as this makes analysing data in a specific domain like Finance, Marketing and Operations easier. The communication skills gained during a MBA also makes client servicing easier.

Apart from formal education and degrees, what other skills and certifications are needed? At the lowest level of the data pipeline are the database administrators, people who work on gathering and maintaining data sets. Even though this is not really Data Science, this Data Infrastructure part, where you query data from the web, APIs, and other places, and build a database, comprises useful skills that every Data Scientist should be able to use. Tools needed to perform these tasks include Excel, Macros in VBA, SQL, noSQL, Hadoop, MapReduce, Mahout, Hive and Pig.

Once you have collected and cleaned your data, you will need to perform exploratory techniques on it, like running plots, descriptive statistics and other visualisations, to summarise your data. This helps inform your hypotheses and should be performed before more complex statistical modelling. Finally, you can create and use statistical software and algorithms to model your data so you can make inferences that answer your questions. The most common models used are summarisation tools used to develop concepts,

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It also helps to build a network of other Data Scientists to keep abreast of developments in the field. Some popular Data scientists to follow on Twitter are, @johnmyleswhite, @hbmason, @pvanimpe and @DiegoKuonen.